AI or Interest Rates?

Parker Buszka · November 2026

> *A note before you read: this is a side project I did in my free time, out of genuine curiosity. I'm not an economist or a labor-market specialist, and it hasn't been peer reviewed. I kept hearing confident claims about AI and jobs and wanted to see what public data, and what I know about causal inference, could actually show. I've tried to be upfront about where it falls short. If you find something I got wrong, I'd like to hear it.*

This is the version for general readers. A technical version, AI Exposure and U.S. Employment After 2022, has the same results with every table, equation and reference, and is in the same repository.

The short version

After 2022, tech job postings in the United States fell hard, and ChatGPT arrived right in the middle of the fall. It's tempting to connect the two. The trouble is that three other things happened in the same stretch: interest rates went from near zero to over 5%, companies that had hired heavily during the pandemic started cutting back, and a change in the tax code made software developers more expensive to employ. Any one of those could produce the same slump.

So I didn't ask whether tech jobs fell. They did. I asked a narrower question that the data can actually answer: after ChatGPT, did the kinds of work AI could do more of lose ground relative to the kinds it can't? Through September 2026, here's what public data says.

- **Inside the same industry, the jobs most exposed to AI haven't lost ground.** Comparing the most exposed quarter of occupations with the least exposed quarter in the same sector, the change from 2022 to 2024–26 is +0.3%, with a 95% confidence interval from −5.6% to +6.1%.
- **Across industries, the most exposed ones slid about 5%.** But that test failed a check I set for it in advance, about 40% of the slide disappears once the pandemic hiring boom is accounted for, and the same model finds a similar slide in the dot-com bust, when there was no ChatGPT.
- **Young workers are the open question.** Their estimates lean negative, but depending on reasonable modeling choices they run from about −1% to −11%, and the main one isn't statistically clear.

Under rules I wrote down before running the main tests, the verdict is "can't tell yet." That isn't the same as "AI has no effect," and the rest of this piece explains how I got there and why I trust the numbers as far as I do.

Why I asked this

There's a lot of debate about AI and jobs. Some people expect AI to make a big share of jobs obsolete. Others expect it to look like the steam engine, painful for some workers but ending up creating more work than it destroys. The truth is that nobody knows yet, me included.

The argument I hear most often is simpler: software jobs fell after ChatGPT came out, so AI must be responsible. That argument compares one group with itself over time, before and after. The problem is that it hands AI the credit, or the blame, for everything else that changed in the same months. Statisticians call

this **confounding**: two things move together because a third thing, or several, moved both of them. And in 2022 a lot moved at once.

When	What happened
Tax years from 2022	A tax rule called **Section 174** starts requiring companies to spread the cost of R&D, including software development, over five years instead of deducting it right away. Developers suddenly cost more after tax.
March 2022	The Federal Reserve starts raising interest rates from near zero, reaching 5.25–5.50% by July 2023. Cheap money for growth hiring dries up.
Through 2022	Indeed reports software job postings already falling "as tech companies rein in hiring plans" (Bunker, 2022).
November 30, 2022	ChatGPT is released.
Tax years from 2025	A July 2025 law restores immediate deduction of domestic R&D costs.

Look at the order. The rate hikes, the tax change and the hiring pullback all start before ChatGPT. That rules out a quick answer, but it also helps, because the shocks didn't all land at the same moment. Differences in timing, and in which jobs each shock should hit, are what make the question testable at all.

What this does and doesn't measure

I want to be clear about the boundaries up front, because the debate is much bigger than what any one analysis can settle.

- **The outcome is employment,** meaning how many people hold jobs, across every occupation and private industry. It isn't tech hiring, job postings, layoffs, wages or hours. (One part looks at hires.)
- **The treatment is potential exposure:** tasks a language model *could* speed up. I have no data on whether anyone actually uses AI at work.
- **Every comparison is relative.** I compare more exposed work with less exposed work. If AI cut jobs everywhere equally, these designs couldn't see it, and they can't say how many jobs AI added or removed in total.
- **The window is short:** the United States, comparing 2024–26 with 2022, with data through September 2026.

So when I say "no change" below, I mean the relative share of jobs AI could do more of stayed within a stated range. I don't mean AI left tech hiring, or employment in general, untouched.

What others have found

The most widely cited study is "Canaries in the Coal Mine" by Erik Brynjolfsson, Bharat Chandar and Ruyu Chen (2026a) at Stanford. They used payroll records from ADP, one of the largest payroll processors in the country, through June 2026. Across the whole economy they find no sign of widespread job loss from AI. The exception is the youngest workers: people aged 22 to 25 in AI-exposed occupations, whose employment now sits 19% below where it would be if it had kept pace with their less exposed peers. Experienced workers in the same occupations show no comparable gap, and the shortfall comes mostly from fewer hires rather than more firings.

The authors are careful about what that means. They describe the pattern rather than claim AI caused it. It survives controls for interest-rate exposure and remote work, but it shrinks when they control for

education, some of the trends start before generative AI existed, and the gap is larger in their payroll data than in public surveys like the CPS (more on that survey below). An earlier note by the same team (Brynjolfsson et al., 2026b) found that, once they compared workers inside the same firm in the same month, the relative decline showed up clearly only after 2024, and they called the earlier declines "likely (at least partly) due to some combination of other factors, not just AI."

ADP's data is private. So my question was how far you can get on the same question with data anyone can download.

One more study shaped a choice I made. Every analysis like this needs a score for how exposed each job is to AI, and those scores usually come from asking a language model to rate tasks. Michelle Yin, Hoa Vu and Claudia Persico (2026) reran that exercise with several different models and found that the models agreed on as little as 57% of ratings, and that the resulting estimates could differ by a factor of 2.4 depending on which model did the scoring. That's why I don't rely on one rater. I average the GPT-4 ratings with ratings from human annotators.

The data

To test this I needed three things: employment broken down by type of job, a score for how exposed each job is to AI, and some way to account for how sensitive each job is to interest rates. No single dataset has all three, so I combined several. Government statistics come wrapped in acronyms, so here is what each one actually is.

CPS, the Current Population Survey. Every month the Census Bureau, working for the Bureau of Labor Statistics (**BLS**), interviews tens of thousands of households about who's working and in what job. It's the survey behind the monthly unemployment rate. It's the only free monthly source that records a person's occupation, industry and age together, and it goes back far enough to cover the dot-com bust. I used a cleaned-up version from **IPUMS**, a University of Minnesota project that makes census and survey data consistent over time (Flood et al., 2025). Its weaknesses: it's a sample, so a single occupation might have only a few hundred respondents a month (software developers included), and the government changed its occupation codes in 2003 and again in 2020. The individual records also can't be shared, so anyone repeating this needs their own free IPUMS account.

CES, the Current Employment Statistics survey. This is the other half of the monthly jobs report: instead of asking households, BLS asks employers how many people are on their payroll (U.S. Bureau of Labor Statistics, n.d.). Its samples are large and its industry definitions are consistent back to 1990, using a standard coding system called **NAICS** (the North American Industry Classification System). Its limitation is that it records industries, not occupations. I used 99 detailed private industries; the 94 with complete data cover about 86% of private-sector jobs.

FRED, Federal Reserve Economic Data. A free database run by the Federal Reserve Bank of St. Louis. It's not a survey itself; it's where I downloaded the CES industry series, interest rates and inflation (Federal Reserve Bank of St. Louis, n.d.).

QWI, the Quarterly Workforce Indicators. The Census Bureau builds these from state unemployment-insurance records, which means they count actual hires, by age, industry and region (U.S. Census Bureau, n.d.). Hiring is exactly where the Stanford team found the effect, so this matters. The downside is a lag of about three quarters, and it only knows industry, not occupation.

The pairing is deliberate. The CPS knows occupation and age but has small samples. The QWI has big samples of real hires but no occupations. If both point the same way, that's much stronger evidence than either one alone.

The exposure scores. These come from "GPTs are GPTs" by Tyna Eloundou and colleagues (2024). They broke each occupation into its tasks and had both GPT-4 and human annotators rate whether a language model could speed each task up. I use their **beta** measure, which counts tasks an LLM can speed up directly plus half of the ones it could speed up with extra software built on top. For each occupation I average the GPT-4 and human scores, then rescale so one unit means one **standard deviation**, a typical gap between two occupations. An industry's score is the average across the occupations it employed in 2015–19, before ChatGPT, so the score can't be contaminated by AI itself. Keep in mind that this measures what AI *could* do, not what anyone is doing with it.

Interest-rate sensitivity. Some jobs swing with interest rates more than others, and I didn't want to mistake a rate effect for an AI effect. So for each industry and occupation group I estimated how much its employment moved after surprise changes in Federal Reserve policy, using a published series of those surprises by Michael Bauer and Eric Swanson (2023) and only 1990–2019 data, years before generative AI. I estimated each job's sensitivity to the ordinary business cycle the same way.

Before trusting any of this, I compared it with published figures. For ages 25 to 54, my CPS employment stays within 0.7% of the official BLS series in every month since 2010. For seven of nine occupations I checked, my 2019–25 change lands within 3 percentage points of the official tables. The two that don't, software developers (+29% against +24%) and managers (+17% against +10%), are probably affected by the 2020 occupation-code change.

How I tested it

The core problem is that you can't rerun 2022 without ChatGPT. Every method below is a different way of building a stand-in for that missing world, what economists call the **counterfactual**: what would have happened to exposed jobs if nothing about AI had changed. No method is trustworthy on its own, so I used several whose weaknesses differ. Something that fools one design is unlikely to fool all of them in the same direction. That approach has a name too, **triangulation**.

I did this in two rounds. Round 1 was exploratory. Round 2 was a set of tests I wrote down before running them.

Round 1: build a twin

The simplest idea is to compare tech with a blend of other industries that used to move just like it. If some weighted mix of industries tracked computer systems design closely for years before 2022, that mix is a reasonable guess at what the industry would have done without ChatGPT, and the gap that opens afterward is the estimated effect. The method is called a **synthetic control**, and it comes from Alberto Abadie, Alexis Diamond and Jens Hainmueller (2010).

My first attempt used computer occupations in the CPS, and it failed a basic test. I fit the twin on earlier years and asked it to predict a few later years it hadn't seen, a **hold-out test**, and its errors were about five times larger than during fitting. The occupation data was too noisy. I kept that as a documented failure and moved to the computer systems design industry in the CES, where the twin fit well (0.6% error on the

held-out years). The biggest ingredient in the blend turned out to be home health care, at 31% of the weight.

By 2026 the industry sits 13% below its twin, and the gap opened in 2023–24. That timing fits AI. But I then ran the same procedure on each comparison industry, pretending in turn that it was the one hit by AI, and plenty of them drifted that far from their own twins by chance: the **placebo p-value** is 0.29. With a single industry, only effects of roughly 16% to 20% stand out from that noise. Inconclusive.

Round 1: pool many twins

To get more power, I grouped industries: the 20 most exposed (top quarter on both the GPT-4 and human ratings) against 43 low-exposure ones (below the median on both). The method is **synthetic difference-in-differences**, or SDiD, from Dmitry Arkhangelsky, Susan Athey and colleagues (2021). It crosses the twin idea with the classic **difference-in-differences**, which compares how much the treated group changed from before to after against how much the comparison group changed over the same stretch. SDiD adds two sets of weights:

$$\hat{\tau} = \left(\bar{Y}_{\text{exp, after}} - \sum_{t} \lambda_t \, \bar{Y}_{\text{exp},t} \right) - \sum_{j} \omega_j \left(\bar{Y}_{j,\text{after}} - \sum_{t} \lambda_t \, Y_{j,t} \right)$$

Read the left side as "tau-hat," the estimated effect. Each Y is log employment, so differences between them are roughly percentage changes. The first bracket is how far the exposed group moved from its "before" level to its average after 2022. The second bracket is the same thing for each comparison industry j. The **unit weights**, omega (ω), decide how much each comparison industry counts; they're chosen so the weighted comparison group tracks the exposed group before 2022. The **time weights**, lambda (λ), decide which earlier quarters count as "before"; they're chosen so the before period predicts the after period as well as possible. The effect is the difference between the two brackets.

Those time weights taught me something. Left to choose for themselves, they put 81% of their weight on the last quarter of 2021, the peak of the pandemic hiring boom. Measured from that peak, exposed industries looked like they'd fallen 3.9%, a statistically significant decline. But that was mostly the boom unwinding. Exposed industries stood about 5.8% above their normal path by 2022 and drifted back to about −3.9% by 2026. Measured from 2019 instead, before the pandemic, the estimate is +0.7%, with a 95% confidence interval from −2.9% to +4.3%.

I chose 2019 as the baseline, and I also chose to drop 2020 through mid-2021, when COVID hit in-person, low-exposure industries for reasons unrelated to AI. Both choices came after I'd seen the data, and the baseline change moved the answer toward zero. That's exactly the kind of decision that can steer a result without anyone meaning to, which is why I wrote down Round 2 before running it.

Round 2, Test A: more exposure, bigger change?

Splitting industries into high and low throws away everything in the middle. Test A uses all 94 industries and treats exposure as a dose: if AI is behind a decline, more exposed industries should have fallen more. This is an **event study** with a continuous treatment:

$$\log E_{it} = \alpha_i + \delta_t + \sum_{k \neq 2019} \theta_k \, X_i \, \mathbf{1}[\text{year}(t) = k] + \sum_{k \neq 2019} \left(\phi_k R_i + \psi_k C_i \right) \mathbf{1}[\text{year}(t) = k] + \varepsilon_{it}$$

It looks like a lot, so take it one piece at a time.

- E is employment in industry i in quarter t. Taking the log means a change of 0.05 is roughly a 5% change.
- Alpha-i (α) is each industry's own normal level, so a huge industry and a tiny one can be compared on the same footing. Delta-t (δ) is anything that hit every industry in quarter t: a recession, a rate hike, the general mood after ChatGPT. Both are called **fixed effects**, and delta is why this design is relative. Anything that moved all industries together gets absorbed, so the test can only see how exposed industries did compared with the rest.
- X is the industry's exposure score, in standard deviations.
- The bold 1 is an **indicator**: it equals 1 in year k and 0 otherwise. Multiplying X by it lets exposure matter differently in each year.
- Theta-k (θ) is the number that matters. It measures how far apart, in employment, two industries one standard deviation apart in exposure were in year k, compared with how far apart they were in 2019.
- R and C are each industry's sensitivity to interest rates and to the business cycle, with their own yearly coefficients (phi and psi), so a rate-sensitive industry can follow its own path without that being mistaken for AI.
- Epsilon (ε) is everything else, the noise.

The headline number compares the AI years with 2022, a year that had rate hikes but almost no AI adoption:

$$\Delta = \tfrac{1}{3}\left(\theta_{2024} + \theta_{2025} + \theta_{2026}\right) - \theta_{2022}$$

Then I multiply by 2.52, the distance in standard deviations between the average industry in the most exposed quarter and the average one in the least exposed quarter, so the result reads as "most exposed versus least exposed."

For this to mean anything, one thing has to be true: without AI, more and less exposed industries would have kept moving the same way relative to each other after 2022. That's the **parallel trends** assumption, and every difference-in-differences design rests on it. The main threat is some other shock that hit exposed industries from 2023 on. And there's a wrinkle I'll come back to: exposed industries were at their post-2019 high in 2022, so comparing with 2022 can pick up the boom unwinding.

Round 2, Test B: same industry, different jobs

Industry totals can hide a lot. An accounting firm could replace junior bookkeepers with software while hiring more salespeople, and its headcount wouldn't move. So Test B compares occupations inside the

same sector: 576 cells, each one an occupation group within a sector (108 occupation groups across 14 sectors), measured yearly from 2011.

$$\log E_{cjt} = \alpha_c + \delta_{jt} + \sum_{k \neq 2019} \theta_k \, X_c \, \mathbf{1}[t = k] + \sum_{k \neq 2019} \psi_k \, C_c \, \mathbf{1}[t = k] + \varepsilon_{cjt}$$

Here c is a cell and j is its sector. Most of the pieces mean the same as before. The one that matters is delta-jt: a separate fixed effect for every sector in every year. Anything that hit a whole sector, such as a rate shock to one sector or the end of a hiring boom in another, gets absorbed, because the comparison is always between occupations in the same sector in the same year. That's what makes Test B the strongest test here. What it can't absorb is a shock that hits some occupations in a sector and not others.

A cell's employment is the CPS survey weights added up across employed respondents and divided by the months surveyed that year. A cell has to have at least 30 respondent-months in every year to be included, cells count in proportion to their 2015–19 employment, and uncertainty is measured allowing each occupation group's errors to be correlated over time.

There's also an age version, which splits each cell into young workers (22 to 27) and older workers (35 to 65) and asks whether the exposure gap got worse for the young. Comparing across exposure, age and time at once is a **triple difference**, and it's the closest I can get to the Stanford comparison with public data. It has a blind spot worth knowing: young workers enter an occupation by being hired and leave the age band as they get older, so anything that slows hiring in an occupation pushes this estimate down, whether or not AI is involved.

Checking whether the tests can tell

Two more tools keep the tests honest.

The first is a **placebo**. The dot-com bust was a tech crash with no generative AI in it, so I ran Test A and SDiD on 1996–2003, with 2001 standing in for 2022. If the same pattern shows up when AI couldn't have caused it, the 2022 pattern needs some other explanation. It's an imperfect twin, since the Federal Reserve was cutting rates in 2001 and raising them in 2022. I also ran fake dates, pretending ChatGPT arrived in 2016 or 2017, where a good design should find nothing.

The second is an **injection test**. A result of "no change" only means something if the method could have seen a change. So I took the real data, planted a fake decline of known size, reran the exact same pipeline, and counted how often it found the plant. That's a direct measure of **statistical power**. Round 2 plants a gradual ramp starting in 2023, since real adoption is gradual. The rule I set in advance: if a design can't reliably detect a decline of 10%, its "no change" counts as uninformative.

The rules I set before looking

Round 1 taught me how easy it is to steer an answer without meaning to. Every choice made after seeing the data (which baseline, which years to drop, which comparison group) is a fork in the road, and with enough forks you can reach almost any conclusion without ever doing anything that feels dishonest. The statisticians Andrew Gelman and Eric Loken (2013) call this the **garden of forking paths**.

The standard defense is a **pre-registration**: write down the questions, the equations, the checks and what each possible result would mean, before running the tests. I did that before Round 2 and recorded a fingerprint of the file (its SHA-256 hash, listed in the repository), so any later edit would show. It isn't a perfect pre-registration, and I should say why. I had already looked at all three datasets in Round 1, and the young-worker comparison uses the same survey where Round 1 had already shown a decline.

The plan also fixed how I'd judge each number.

- **Confidence intervals and clustering.** Each estimate comes with a 95% confidence interval: a range that should contain the true effect 95% of the time if the method is sound. An industry's employment this quarter isn't independent of last quarter's, so the intervals allow for that by grouping each unit's observations together, which is called **clustering**.
- **Three p-values.** A **p-value** is how surprising a result this large would be if there were truly no effect. I report three versions that rest on different assumptions: the standard formula, a **wild cluster bootstrap** (which randomly flips the signs of each industry's errors 999 times to see how much the estimate bounces around), and **randomization inference** (which shuffles exposure scores across industries 1,000 times and asks how often a fake result beats the real one). When all three agree, the answer isn't an artifact of one method.
- **Two main tests, one correction.** Tests A and B are the two primary questions. Running two tests roughly doubles the chance of a fluke, so the smaller p-value has to clear 0.025 rather than the usual 0.05, a rule called the **Holm correction** (Holm, 1979).
- **The probability the change is negative.** I also report, in plain terms, how likely it is that the true change is below zero. It's easier to read than a p-value, though it speaks to the direction of a change, not its cause.
- **The pre-trend check.** If parallel trends holds, exposed and less exposed work should have been moving together before the event. So I test whether the gaps before 2019 are jointly zero, and a design has to clear $p > 0.10$ to pass. It's an imperfect test, because a slow drift can slip through, but a design that fails it shouldn't be trusted.

And I wrote down the four possible verdicts in advance.

Verdict	What it takes
1. Evidence AI is reducing employment in exposed work	A main estimate that's negative and survives the Holm correction, from a design that passes its checks, isn't already half there in 2022, differs from the dot-com bust or passes a fake-date check, holds in most reasonable variations, and has support from at least two other angles
2. Evidence AI is increasing employment in exposed work	The same, with the sign flipped
3. No effect larger than X so far	Both main tests pass their checks and neither is significant
4. Can't tell yet	Anything else, including a significant result from a design that fails its checks

Those rules turned out to matter. Test A's verdict changed twice as I found and fixed my own mistakes. An outside review caught that I had built the industry scores from the wrong years of occupation data, and a later audit of my own found two errors in how industries were matched to occupations. Each fix flipped the pre-trend check: fail, then pass, then fail again ($p = 0.078$, then 0.107, then 0.073).

Across the six versions of the industry score I ran, the pre-trend p-value lands anywhere from 0.067 to 0.107, and only one of them passes. The estimate, meanwhile, barely moves, staying between −3.7% and −5.0%. When a yes-or-no test sits right on its threshold, the label carries less information than the number. So read the estimates first, and treat the verdict as fragile.

What I found

Unless I say otherwise, everything below compares the most exposed quarter of work with the least exposed quarter, using data through September 2026 (2025 for hires), and every number is relative to less exposed work. Here are the four main results side by side before I go through them.

Question	Test	Estimate	95% range	Passed its pre-trend check?
Did exposed occupations lose ground inside the same sector?	Test B, CPS	+0.3%	−5.6% to +6.1%	Yes
Did more exposed industries do worse?	Test A, CES	−5.0%	−9.0% to −0.9%	No
Did hiring of 22–24 year olds fall in exposed industries?	SDiD, QWI	−9.3%	−18.3% to −0.2%	Yes, but the gap opened before ChatGPT
Did young workers fall behind older ones in the same jobs?	Triple difference, CPS	−6.3%	−15.5% to +2.9%	Yes

Across industries: a slide that's hard to pin on AI

Test A finds that the most exposed industries fell 5.0% relative to the least exposed between 2022 and 2024–26, with a 95% range from −9.0% to −0.9%. All three p-values agree it's unlikely to be chance (0.017, 0.016 and 0.010), and the probability that the true change is negative is 99%. The decline isn't spread smoothly by exposure. It sits in the more exposed half: compared with the least exposed fifth of industries, the next four fifths come in at +1.7%, +0.4%, −4.0% and −3.9%. Taken alone, that's the kind of pattern AI would produce, starting in 2023. I don't treat it as evidence of an AI effect, but I can't rule one out either, and four things stand in the way.

1. **It fails the pre-trend check** (p = 0.073). Exposed industries were already gaining on the rest before 2019, which means the true shortfall could be bigger than 5.0%, not smaller. Under my rules, though, a design that fails this check doesn't get interpreted.
2. **It would have survived the multiple-testing correction.** Its p-value of 0.017 is under the 0.025 bar. So the reason it isn't interpreted is the pre-trend check, not the noise.
3. **About 40% of it runs through the pandemic boom.** Exposed industries grew well above their usual path in 2020–22. When I control for how far each industry sat above its 2015–19 trend in 2022, the estimate shrinks to −2.9% (−7.1% to +1.2%), and that version passes the pre-trend check. The control could also soak up a real AI effect, if AI is what ended the boom, so the difference between the two is best read as the part this design can't assign to either story.
4. **The dot-com bust looks similar.** The same model, run on 1996–2003, gives −3.9% (−6.6% to −1.1%), with no ChatGPT anywhere. That placebo fails its own pre-trend check (p = 0.0004), so it's suggestive rather than decisive.

Three other checks sit beside this. If I extend the pre-2019 trend in a straight line instead of assuming the groups would have stayed parallel, the shortfall grows to −6.1% (−9.5% to −2.7%), which shows the

industry conclusion depends on how you treat that earlier drift. Pretending ChatGPT arrived in 2017 gives −0.1%, which is what a sound design should show. And weighting industries by how many people they employ, so the result describes the typical job rather than the typical industry, gives −5.5% (−9.0% to −2.1%). That weighted version narrowly passes its pre-trend check (p = 0.104), stays between −4.6% and −6.4% when any single industry is dropped, and is the clearest sign of a relative decline anywhere in this study. It also shrinks to −3.1% with the boom control, and the same weighted model shows −4.0% in the dot-com bust.

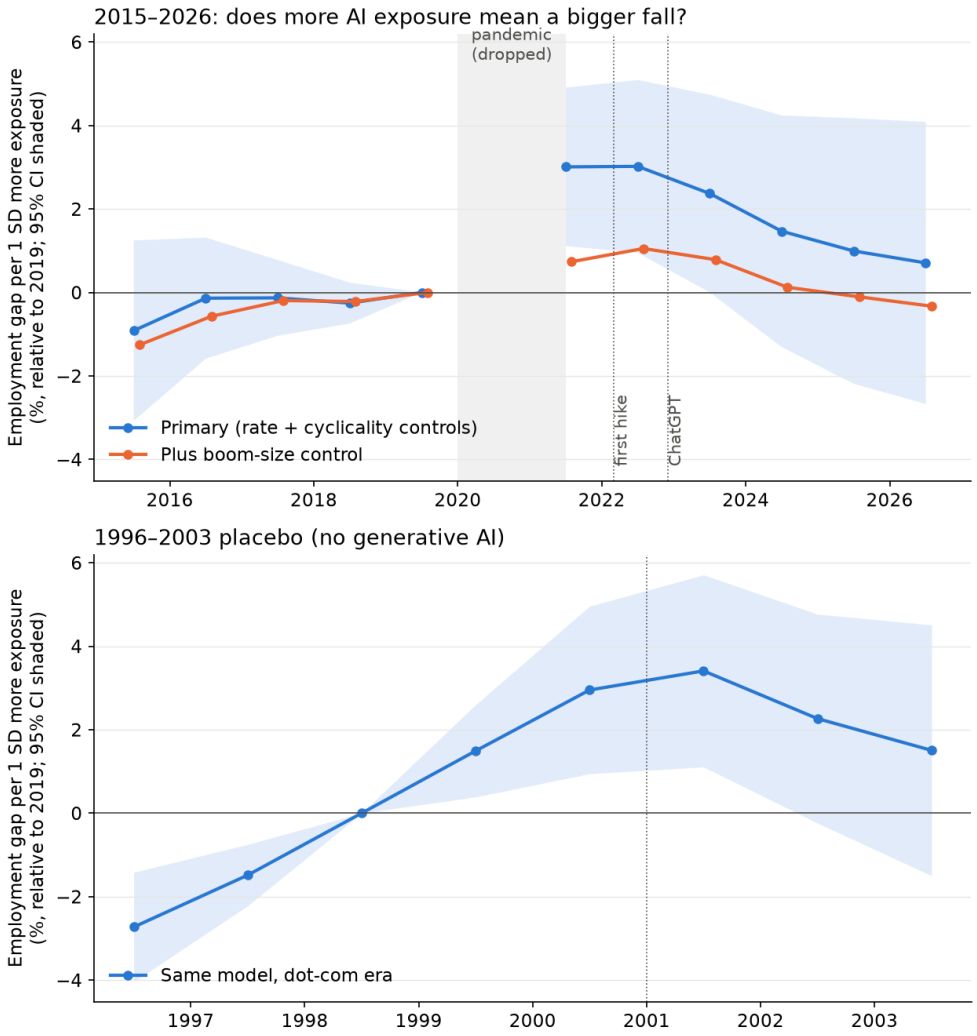


Figure 1. How far apart, in employment, two industries one standard deviation apart in exposure were each year, compared with 2019. Shaded bands are 95% confidence intervals. Top: 2015–2026, with and without the control for the pandemic boom. Bottom: the same model in the dot-com era, when there was no generative AI. Both show exposed industries rising and then sliding back.

The synthetic DiD from Round 1 tells the same story from a different angle. Measured from 2019, the exposed group is +0.7% (−2.9% to +4.3%). It stood about 5.8% above normal in 2022 and drifted back down. And Test A's own coefficients show the exposed industries at their post-2019 high in 2022 (+3.0% per standard deviation of exposure relative to 2019), which is exactly the year it uses as its baseline.

Within industries: no sign of exposed jobs losing ground

This is the most informative result in the study for workers of all ages. Inside the same sector, the most exposed occupations changed by +0.3% relative to the least exposed (−5.6% to +6.1%). All three p-values are around 0.9, and the probability that the true change is negative is 47%, which is a coin flip. The pre-trend check passes (p = 0.23).

The result holds up when I change things around it.

Version	Estimate	95% range
As planned	+0.3%	−5.6% to +6.1%
Without computer occupations	−0.4%	−6.6% to +5.8%
Without the information and professional services sector	0.0%	−6.2% to +6.2%
Removing each cell's own straight-line trend	−1.9%	−8.0% to +4.2%
Fake ChatGPT in 2016	+0.1%	−3.6% to +3.9%
Adding each occupation's share of college graduates (exploratory)	−0.5%	−8.4% to +7.4%

A "no change" result is only worth something if the test could have seen a change, so here is how sensitive it is. The interval rules out an average shift larger than about 6% in either direction. If a decline built up gradually, though, it would need to reach about 13% by the end of the period for this test to catch it 80% of the time. I'd call this a **bounded null**: no sign of an effect, with a clear limit on how large an average effect could be hiding. The education check matters too, because exposed occupations tend to employ more college graduates (the two correlate at 0.65). Adding that control leaves the estimate near zero but widens the range.

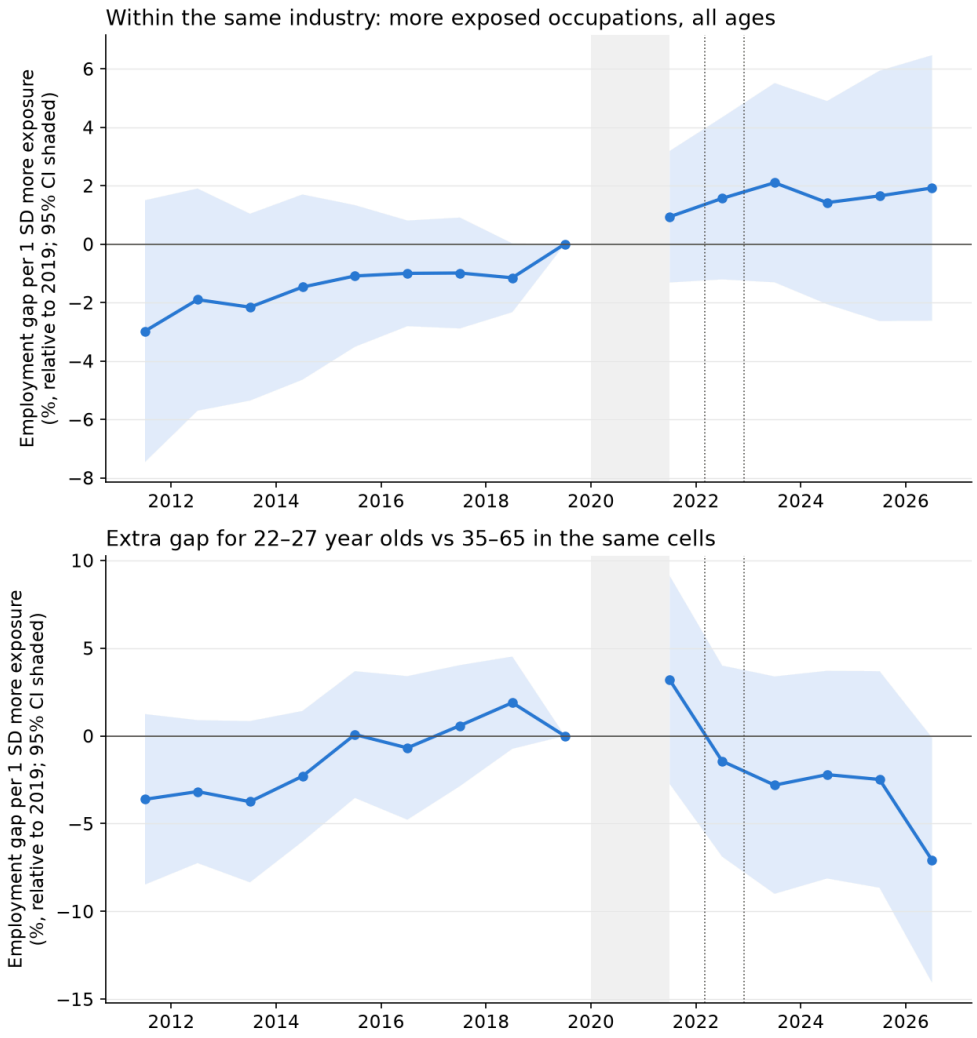


Figure 2. Top: more exposed occupations against less exposed ones in the same sector, all ages, relative to 2019. After 2022 the line sits at about +1% to +2% per standard deviation, and the band includes zero throughout. Bottom: the extra gap for 22- to 27-year-olds against 35- to 65-year-olds in the same cells. It drifts down after 2022 and drops in 2026, but the band is wide and 2026 is a partial year.

This doesn't contradict the Stanford result, because it covers workers of all ages, and the Stanford effect is about the youngest workers. That's where I looked next.

Young workers: leaning negative, size unclear

If AI is replacing entry-level work, young workers should show it first. My two sources agree on the direction but not on the timing.

In the QWI, hires of 22- to 24-year-olds fell 9.3% relative to hires of 35- to 44-year-olds in exposed industries, compared with less exposed ones (−18.3% to −0.2%). Here "exposed" means above the median on both ratings, because the QWI has too few industries to compare quarters. That's a real gap. The problem is when it opened: in 2020–21, before ChatGPT existed. It then held almost flat, at −10.2%, −8.6%, −8.9% and −9.3% across 2022–25. A model released in late 2022 can't explain a gap that was

already there, although a gap that stays put can't rule out AI preventing a recovery that would otherwise have happened.

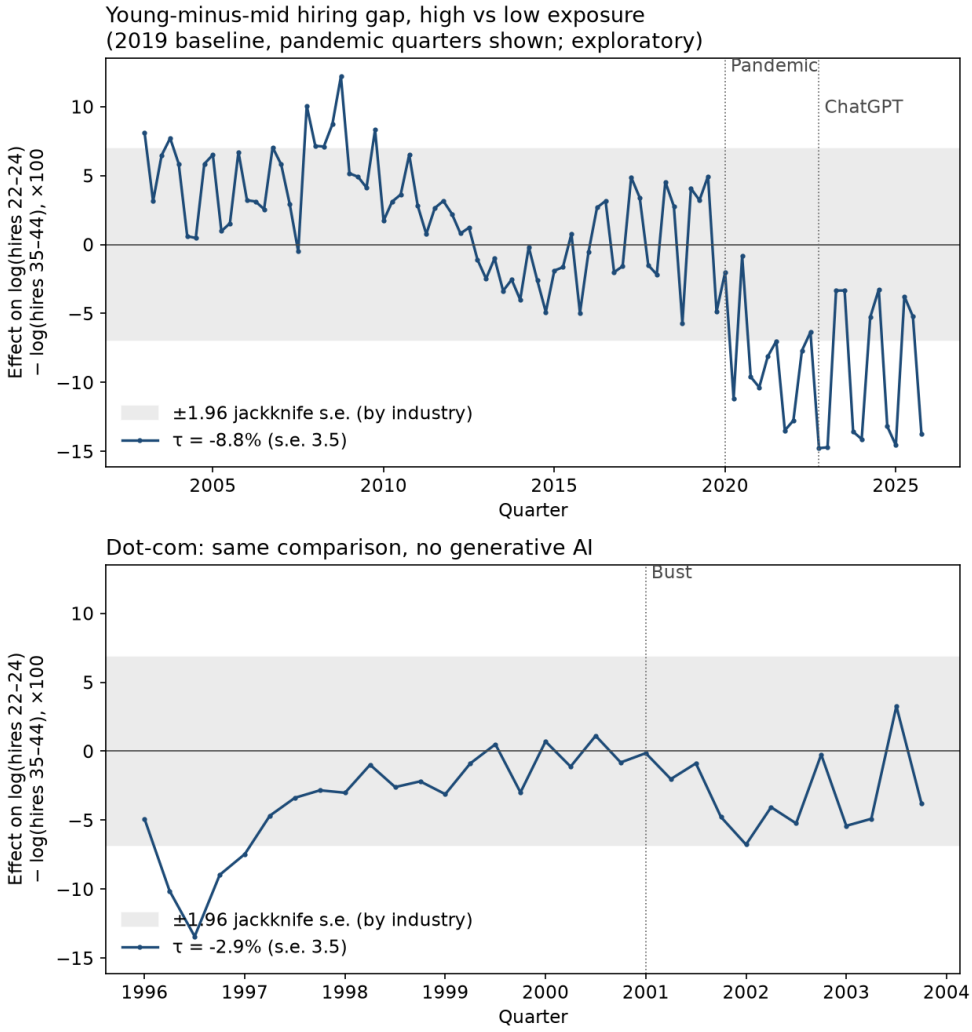


Figure 3. Top: hires of 22- to 24-year-olds minus hires of 35- to 44-year-olds, in exposed industries against less exposed ones. This version keeps the pandemic quarters so you can see when the gap opened (here −8.8%; the main estimate, which drops them, is −9.3%). The gap drops in 2020–21, before ChatGPT. Bottom: the same comparison around the dot-com bust.

In the CPS, the triple difference finds young workers fell 6.3% relative to older workers in the same occupation and sector (−15.5% to +2.9%, p = 0.18), with a 91% probability that the true change is negative. That's the same direction as the Stanford study. But most of the move comes in 2026: the yearly gaps per standard deviation of exposure are −1.4%, −2.8%, −2.2%, −2.5% and then −7.1% across 2022–26, and 2026 is a partial year of noisy survey data. Drop 2026 and the estimate falls to −2.3%.

The estimate is also sensitive to reasonable modeling choices. None of the checks below were planned in advance, and none are corrected for running several tests at once.

Version	Estimate	95% range
As planned	−6.3%	−15.5% to +2.9%
Excluding 2026	−2.3%	−11.9% to +7.3%
Adding age-specific controls for rates and the business cycle	−4.7%	−14.1% to +4.6%
Using January–September in every year	−2.7%	−13.6% to +8.2%
Measured from 2019 instead of 2022	−9.9%	−25.3% to +5.5%
Removing each cell's own straight-line trend	−11.4%	−21.3% to −1.4%
Comparing young and older workers only within the same cells	−0.9%	−10.3% to +8.5%

The last row matters most. The planned version compares 167 young cells with 484 older cells, mostly different ones, so part of its gap comes from which cells end up compared. When young and older workers are compared only inside the same occupation and sector, which absorbs everything that hit that cell in a given year, the estimate falls to −0.9%. I keep −6.3% as the reported number, because it's what I planned, and picking among these after seeing them would be the forking-paths problem all over again. The fair reading is that the young-worker estimates lean negative, but these data can't pin down how big the effect is.

The dot-com bust doesn't help much here. Neither source shows a clear tilt against young workers then, but both intervals are too wide to rule one out.

What else could explain it

The timing favors the other explanations for the broad industry decline. The rate hikes, the end of the boom and the tax change all came before ChatGPT, and the industry pattern looks like a boom unwinding. Test B, which absorbs anything that hit a whole sector, finds no decline. So the post-2022 pattern doesn't need AI to explain it. What this study can't do is say how much each alternative contributed, and it doesn't test the tax change directly.

How much the tests could see

Test	Planted decline	How often it was found
Test A, across industries	Gradual, reaching 10% by late 2026	94% of the time; reliably detects about 8%
Test B, within industries	Gradual, reaching 10%	88–90% on shuffled data; about 13% using its real uncertainty
SDiD, industries	Sudden 10% drop in 2023	100%, dated correctly
SDiD, young workers (QWI and CPS)	Sudden 15% drop in 2023	About 40%; needs about 20–24%
Triple difference, CPS	Not tested directly	Needs about 13%, from its uncertainty alone

The industry tests can see moderate effects. The young-worker tests can't, which means "no clear widening for young workers since ChatGPT" is weak evidence either way. That remains the main open question.

The verdict

Under the rules I set in advance, the result is Verdict 4: can't tell yet. Test B passes its checks and shows no meaningful change. Test A fails its pre-trend check, and a design that fails that check doesn't get interpreted, even though its estimate would otherwise clear the multiple-testing bar. Four points matter more than the label:

- **The label is fragile.** Small, defensible choices in how industries get their exposure scores decide whether Test A passes. The estimate barely moves.
- **The industry decline is hard to attribute.** About 40% of it goes away when industries with similar pandemic booms are compared, and the same model shows a similar decline in the dot-com bust.
- **Within sectors, there's no sign of a large average change.**
- **This is not evidence of no effect.** Test A's estimate is negative with a range that excludes zero, and the young-worker estimates lean negative, even if they aren't statistically clear and one comparison puts them near zero.

How I checked my work

A mistake in code or data doesn't announce itself. It produces a number that's precise, plausible and wrong. So before publishing, I tried to break the analysis rather than confirm it.

What I checked	Against what	What happened
The estimators	Simulated data with an effect I planted, so I knew the right answer	Each one found the planted effect
The event studies	A separate refit in pyfixest, a standard econometrics package, using none of my code	Matched to four decimal places, and caught a bug (below)
The bootstrap p-values	The wildboottest package	Agreed within the expected random variation
The synthetic DiD	The synthdid package	Estimates within 0.02 percentage points
The CPS data	The raw IPUMS file and official BLS totals	Within 0.7% of BLS for ages 25–54, every month since 2010
The CES and QWI data	FRED and the Census Bureau's own data service	Spot checks matched exactly
The industry exposure scores	A printed list of what each industry was matched to, read line by line	Found two matching errors
Every reported number	The saved results, through an automated test and a line-by-line check	All match
The whole project	A fresh copy, a fresh software setup, and the raw data, rerun from scratch	All 6,861 saved numbers came back identical

The checks found real problems. The independent refit exposed a bug that had silently left out part of the young-worker model, and fixing it moved that estimate from −6.8% to −6.3% and its p-value from 0.087 to 0.18. Reading the industry matches line by line found the two errors that flipped Test A's verdict. An outside review, checked against the code, found the occupation-mix mistake, and comparing my plan with my code found a pre-registered check that hadn't been run. Each of those is fixed above, and the repository lists every correction with its date.

There's a limit to what this proves. These checks show the numbers are computed the way I describe and can be reproduced. They can't show that the assumptions behind the designs hold, and they were run by me rather than by someone independent.

What this can't tell us

The claim this study supports is narrow on purpose. Here's what stands between these results and a bigger conclusion.

- **It's a side project.** No peer review and no specialist co-authors. I used generative AI to help write code and documents, so treat that source with the same scrutiny as any other.
- **Exposure is potential, not use.** I don't observe who actually uses AI. The scores are a 2023 snapshot, and raters disagree: GPT-4 and the human annotators agree on only 65.6% of individual tasks (Eloundou et al., 2023), although by my calculation their scores correlate at 0.89 across occupations. The errors in a measure like this aren't random noise, so I can't assume the true effect is bigger than what I found.
- **Everything is relative.** Even the least exposed industries score about 0.27 on the GPT-4 rating. If AI touches all work, or moves work from exposed jobs into the comparison jobs, these designs will misjudge it or miss it.
- **Some shocks slip through.** Test B can't absorb shocks that hit some occupations in a sector and not others: the tax change for R&D staff, education, return-to-office mandates, immigration policy, tariffs.
- **The boom and AI can look alike.** "The boom unwound" and "AI kept the boom from lasting" predict the same employment path. Output data would separate them: if output held up while jobs fell, that points to a labor-saving technology; if both fell, it points to weaker demand.
- **The pre-trends.** Test A fails its pre-trend check, and its answer moves from −5.0% to −6.1% when the earlier trend is extended. Comparing industries at different levels of a continuous dose also needs a stronger version of the parallel-trends assumption than a simple before-and-after comparison does (Callaway et al., 2024).
- **The newest numbers are the least settled.** CPS samples are small, occupation codes changed in 2020, CES gets revised every year, and 2026 covers only January through September.
- **The window is short and the young-worker tests are weak.** An effect that's still building may not show up yet, and the age designs can miss declines smaller than about 13–24%.
- **Headcounts only.** I didn't look at hours, wages or job quality, and 14 broad sectors can hide a lot of change inside individual firms.
- **Some choices came after seeing data.** The 2019 baseline, the pandemic exclusion and the corrections to Test A's exposure scores all did. Each is documented, and the original versions are reported.

Where this leaves us

This doesn't show whether AI caused the slump in tech hiring, because it doesn't measure tech hiring directly. What it measures is employment across the whole economy, compared by how exposed each kind of work is to AI. On that question, inside the same sector, the most exposed occupations look about the same as the least exposed. Across industries, exposed work slid about 5%, but that test fails its own

check, part of the slide may be the pandemic boom unwinding, and the dot-com bust produced something similar with no AI at all.

I'm not going to argue that AI has no effect on jobs. Relative comparisons of potential exposure over a few years can't support that. What I will argue is narrower: through September 2026, public data shows no large average loss, timed to AI, in the share of jobs AI could do more of, and it can't rule out a larger decline building gradually. The place to keep watching is young workers. Their estimates lean negative, one careful comparison puts them near zero, and that's exactly where these tests are weakest. The American Community Survey, with roughly 50 times the CPS sample, would be the natural way to settle it.

The code and the sources

- **The code, results, verification audit and every correction:** github.com/pbuszka1/ai-or-interest-rates

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